

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Relevance to Application (1.25)	Use of Cloud Concepts (1.25)	Practical Insight (1)	Completeness (0.75)	Clarity (0.75)	TOTAL (5)
Weightage	25%	25%	20%	15%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Application Based Questions

1. How does cloud federation improve resource sharing and availability across datacenters?
2. What are the key security issues in virtualization, and how can they be mitigated?
3. How does a software-defined datacenter enhance scalability and management?
4. How can identity and access control ensure data privacy in cloud-based healthcare systems?
5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

1. How does cloud federation improve resource sharing and availability across datacenters?

Solution

Cloud federation is the integration of multiple cloud datacenters that work together as a single virtual cloud environment. It enables organizations to share computing resources such as virtual machines, storage, and network services across geographically distributed datacenters.

Resource sharing is improved because workloads can be dynamically distributed between datacenters based on resource availability. If one datacenter becomes overloaded, additional workloads can be shifted to another datacenter with available resources, ensuring efficient utilization of infrastructure.

Cloud federation also improves availability by providing redundancy and failover capabilities. In the event of hardware failures, power outages, or network disruptions in one datacenter, services can automatically continue running from another datacenter with minimal downtime. Load balancing distributes user requests across multiple locations, reducing latency and preventing server overload. Overall, cloud federation enhances scalability, reliability, disaster recovery, business continuity, and efficient resource utilization.

2. What are the key security issues in virtualization, and how can they be mitigated?

Solution

Virtualization introduces several security challenges because multiple virtual machines (VMs) share the same physical hardware through a hypervisor. Common security issues include hypervisor attacks, VM escape, unauthorized access, data leakage between virtual machines, insecure VM images, and resource exhaustion caused by malicious or compromised virtual machines.

These risks can be mitigated by using secure and regularly updated hypervisors, implementing strong Identity and Access Management (IAM), Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC). Network isolation using VLANs or Software Defined Networking (SDN) prevents unauthorized communication between virtual machines. Encryption should be used for data at rest and in transit, while continuous monitoring, intrusion detection systems (IDS), firewalls, vulnerability scanning, and regular security patching help detect and prevent attacks. Backup and disaster recovery strategies further improve security and system resilience.

3. How does a software-defined datacenter enhance scalability and management?

Solution

A **Software-Defined Datacenter (SDDC)** virtualizes and manages computing, storage, networking, and security resources through software instead of manual hardware configuration. All infrastructure components are centrally controlled using management software and automation tools.

Scalability is enhanced because resources such as virtual machines, storage, and networks can be automatically provisioned or expanded according to workload demand without replacing physical hardware. Organizations can quickly add or remove resources, enabling rapid deployment of applications.

Management becomes simpler through centralized monitoring, automated provisioning, policy-based resource allocation, and orchestration. Administrators can manage the entire datacenter from a single interface, reducing manual effort and operational costs. SDDCs also improve resource utilization, increase flexibility, simplify disaster recovery, and support modern cloud environments by enabling automation and faster service delivery.

4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

Solution

Cloud-based healthcare systems store sensitive patient information, making strong identity management and access control essential for protecting data privacy. Identity management verifies the identity of users such as doctors, nurses, administrators, and patients before granting access to healthcare applications.

Role-Based Access Control (RBAC) ensures that users can only access information relevant to their responsibilities. For example, doctors can access patient medical records, nurses can update treatment information, administrators can manage system settings, and patients can view only their own health records.

Multi-Factor Authentication (MFA) adds an extra layer of security by requiring passwords along with OTPs or biometric verification. Sensitive medical data should be encrypted during transmission using TLS and stored securely using AES-256 encryption. Audit logs should record all user activities to detect unauthorized access, while regular security monitoring and compliance with healthcare regulations such as HIPAA help maintain confidentiality, integrity, and privacy of patient data.

5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Solution

Virtualization enables multiple virtual machines (VMs) to run on a single physical server by sharing hardware resources through a **hypervisor**. The hypervisor manages CPU, memory, storage, and network allocation for each virtual machine while ensuring isolation between them.

Scalability is achieved because additional virtual machines can be created or removed quickly based on application demand without purchasing new physical servers. Features such as live migration allow workloads to move between physical hosts without downtime, ensuring continuous service availability.

Resource efficiency is improved by maximizing hardware utilization through resource pooling. Instead of dedicating one physical server to a single application, multiple applications can share the same server while receiving allocated resources according to their requirements. Hypervisors also support load balancing, dynamic resource allocation, fault isolation, backup, and disaster recovery, resulting in lower infrastructure costs, better performance, higher availability, and efficient cloud resource management.